

Lab 2 — Summarising one variable, then two

Department: Data Science and Analytics

Semester: 2026–2027 Fall

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Session: Week 2 · w/c 5 October 2026

Paired lecture: DSA 101, week 2

1. What you will produce

A runnable Colab notebook, pushed to your dsa107-labs repository. In it you summarise one categorical variable, count two categorical variables together, turn those counts into rates, summarise one quantitative variable, and draw three labelled figures.

2. Objectives

By the end of this session you should be able to:

- Decide whether a variable is categorical or quantitative, even when its stored type is misleading
- Summarise one categorical variable as counts and as proportions
- Cross-tabulate two categorical variables and read a single cell out loud
- Turn a table of counts into a table of rates, and say which question each table answers
- Describe a quantitative variable with a histogram, a centre and a spread

3. Background

Last week you loaded a dataset and looked at it. This week you summarise it. The order is always the same: summarise first, then draw. Which summary you may use depends on the kind of variable. A categorical variable is summarised by counting each category. A quantitative variable has too many different values to count, so its values are grouped into ranges and the ranges are counted instead — that is a histogram — and then you report a centre and a spread.

The hard idea of the week is the denominator. One table of counts becomes a different table depending on what you divide by, and the denominator you choose is the question you are asking. A count tells you how many; only a rate lets you compare groups of different sizes. The dataset is a version of the Titanic passenger list. It is not the same file as the one used in the lecture, so your numbers will not match your lecture notes. That is intended: the method has to work on a file you have not seen before.

Data used

Dataset	Where it comes from
Titanic passengers (891 rows, 15 columns)	https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv

4. Tasks

Work through the starter notebook. It contains the detailed steps; this list is what you are accountable for.

1. Open the starter notebook in Colab and run the setup cell.
2. Load the Titanic dataset from the URL given, then report its shape, its column types, and the number of missing values in each column.
3. Summarise passenger class as counts and as proportions, and draw a labelled bar chart of the counts.
4. Cross-tabulate passenger class against survival, and report the marginal total for each class.
5. Turn the same table into a conditional distribution, and draw a labelled bar chart of the rates.
6. Draw a labelled histogram of age, then report the mean, median and standard deviation of age, and the mean, median and largest value of fare.
7. Answer the four questions in the notebook, each with code and a written sentence.
8. Push your completed notebook to your dsa107-labs repository.

5. Deliverables

- The completed notebook, saved as Lab02_Firstname_Lastname.ipynb
- A lab report of 2–4 pages, saved as Lab02_Firstname_Lastname.pdf
- If you used an AI assistant, the log required by the course AI policy

Report structure

1. Title page — course, lab number, report title, your name, student ID and date.
2. Objective — what the lab set out to do, in your own words.
3. Data — what you used, where it came from, and what you changed.
4. Method — what you did and why, not a transcript of your code.
5. Results — the figures and numbers, each one read in a sentence.
6. Discussion — what it means, and what it does not show.
7. Reflection — what was difficult, and what you would do differently.

6. Grading (25 points total)

Each lab is marked out of 25 points: 10 points for the notebook and code, and 15 points for the lab report.

Code grading (10 points)

Criterion	What earns the marks	Points
Setup and data loading	Notebook runs end to end; dataset loads from the URL	2
Summarising one variable	Counts and proportions correct; the proportions add to 1	2
Counts and rates	Crosstab built correctly, and rows normalised to give rates rather than counts	2

Criterion	What earns the marks	Points
Centre, spread and histogram	Histogram drawn; mean, median and standard deviation reported correctly	1
Figures	All three figures have axis labels and a title	1
Interpretation	Four written answers, each reasoning from the output	2
Total		10

A figure without a written reading of it scores nothing on interpretation, however good the figure is. That is the same standard as the lecture course.

Report grading (15 points)

Criterion	What earns the marks	Points
Title page	Course, lab number, report title, your name, student ID and date	1
Objective	Clear summary of what the lab set out to achieve	2
Data description	Dataset source, structure and preprocessing steps described accurately	3
Methodology	Explanation of the analytical process, tools and techniques used	3
Results	Key findings summarised with tables, graphs or statistics	2
Discussion	Interpretation of results and connection to course concepts	2
Reflection	Challenges faced and lessons learned	2
Total		15

Note: Rubrics may vary by lab — check the handout for each specific assignment.

7. Submission

Submit both files through MS Teams before the stated deadline. Before you submit, restart the kernel and run the notebook from top to bottom: a notebook that only works in the order you happened to write it does not count as running.

Note: Discussing ideas with classmates is encouraged; submitting their code, text or figures as your own is not, and the usual academic integrity rules apply.